

Problem Set: PCA in Genomic Analysis

Principal component analysis (PCA) is widely used in genomics to summarize genetic variation across individuals and reveal population structure.

Scenario

You have toy genotype data from 90 individuals (30 from each of 3 populations) across 200 SNPs. Each genotype is coded as 0, 1, or 2 copies of the reference allele.

Expected output
A PCA scatter plot

Tasks

1. Load or simulate the genotype matrix.
2. Standardize SNPs before analysis.
3. Run PCA and plot PC1 vs PC2.
4. Color samples by population label.
5. Interpret what each point and each axis mean.
6. Which populations are clearly separated?
7. If admixed individuals were added, where might they appear?

Python Code for PCA

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA

np.random.seed(42)
n = 30
p = 200

p1 = np.random.beta(0.8, 0.8, p)
p2 = np.clip(p1 + np.random.normal(0.12, 0.05, p), 0.01, 0.99)
p3 = np.clip(p1 - np.random.normal(0.10, 0.05, p), 0.01, 0.99)

def sim(freqs):
    return np.random.binomial(2, freqs, size=(n, len(freqs)))

X1 = sim(p1)
X2 = sim(p2)
X3 = sim(p3)
X = np.vstack([X1, X2, X3]).astype(float)
labels = np.array(["Pop1"]*n + ["Pop2"]*n + ["Pop3"]*n)

X = (X - X.mean(axis=0)) / X.std(axis=0, ddof=1)
X = np.nan_to_num(X)
pcs = PCA(n_components=2).fit_transform(X)
```

What this code does

- Simulates genotype data for 3 populations
- Standardizes SNP values
- Runs PCA
- Plots PC1 and PC2